

DNS Server Configuration using BIND

1. Domain Configuration

- Domain Name: **ditiss.com**

2. Forward Zone Configuration

- Zone Name: **ditiss.com**

Resource Record:

server200 IN A 192.168.220.130

3. Reverse Zone Configuration

- Network: **192.168.159.0/24**
- Reverse Zone Name: **159.168.192.in-addr.arpa**

PTR Record:

200 IN PTR server200.ditiss.com.

4. Firewall Configuration

- DNS service (port 53) is allowed in the firewall

5. DNS Service Management

- DNS service is restarted to apply changes

6. Verification

- Forward lookup resolves:
 - server200.ditiss.com → 192.168.220.130
- Reverse lookup resolves:
 - 192.168.220.130 → server200.ditiss.com

✓ Outcome

- DNS server successfully resolves both **forward (A record)** and **reverse (PTR record)** queries
- System is fully operational for the domain **ditiss.com**

```
#####
```

```
# DNS SERVER COMPLETE SETUP
```

```
# Using BIND (named)
```

```
#####
```

```
#####
```

```
# 1. INSTALL DNS PACKAGES
```

```
#####
```

```
# Install BIND DNS server and utilities
```

```
dnf install -y bind bind-utils
```

```
#####
```

```
# 2. NETWORK CONFIGURATION
```

```
#####
```

```
# Configure static IP (use nmtui GUI)
```

```
nmtui
```

```
# Verify IP configuration
```

```
ip a
```

```
# Check DNS resolver config
```

```
cat /etc/resolv.conf
```

```
#####
```

```
# 3. MAIN DNS CONFIGURATION FILE
```

```
#####
```

```
# Edit main config
```

```
vim /etc/named.conf
```

Key configurations:

- Listen on server IP

- Allow queries from any

- Define forward & reverse zones

---- IMPORTANT CONFIG ----

options {

listen-on port 53 { 192.168.220.130; }; # DNS server IP

directory "/var/named"; # Zone files

location

allow-query { any; }; # Allow all

clients

recursion yes; # Enable recursion

(lab setup)

};

---- FORWARD ZONE ----

zone "ditiss.com" IN {

type master;

file "frwd.ditiss.com";

};

---- REVERSE ZONE ----

zone "220.168.192.in-addr.arpa" IN {

type master;

file "rvs.ditiss.com";

};

```
#####
```

```
# 4. CREATE ZONE FILES
```

```
#####
```

```
cd /var/named
```

```
#####
```

```
# 4A. FORWARD ZONE FILE
```

```
#####
```

```
vim frwd.ditiss.com
```

```
# Content:
```

```
$TTL 1D
```

```
@ IN SOA server200.ditiss.com. abc.gmail.com. (
```

```
    1      ; serial (increment on change)
```

```
    1D     ; refresh
```

```
    1H     ; retry
```

```
    1W     ; expire
```

```
    3H )   ; minimum
```

```
@ IN NS server200.ditiss.com.
```

```
# A Records (Domain → IP)
```

```
ditiss.com. IN A 192.168.220.130
```

```
www IN A 192.168.220.130
```

```
server200 IN A 192.168.220.130
```

```
#####
```

```
# 4B. REVERSE ZONE FILE
```

```
#####
```

```
vim rvs.ditiss.com
```

```
# Content:
```

```
$TTL 1D
```

```
@ IN SOA server200.ditiss.com. abc.gmail.com. (
```

```
    2      ; serial (must increment)
```

```
    1D
```

```
    1H
```

```
    1W
```

```
    3H )
```

```
@ IN NS server200.ditiss.com.
```

```
# PTR Record (IP → Domain)
```

```
130 IN PTR server200.ditiss.com.
```

```
#####
```

```
# 5. PERMISSIONS (CRITICAL)
```

```
#####
```

```
# Set correct ownership for BIND
```

```
chown root:named frwd.ditiss.com rvs.ditiss.com
```

```
# Set secure permissions
```

```
chmod 640 frwd.ditiss.com rvs.ditiss.com
```

```
#####
```

```
# 6. ENABLE & START DNS SERVICE
```

```
#####
```

```
systemctl enable named
```

```
systemctl restart named
```

```
# Check status
```

```
systemctl status named
```

```
#####
```

```
# 7. FIREWALL CONFIGURATION
```

```
#####
```

```
# Allow DNS service (port 53)
```

```
firewall-cmd --add-service=dns --permanent
```

```
firewall-cmd --reload
```

```
#####
```

```
# 8. VERIFICATION
```

```
#####
```

```
# Forward Lookup (Domain → IP)
```

```
host ditiss.com
```

```
host server200.ditiss.com
```

```
host www.ditiss.com
```

```
# Reverse Lookup (IP → Domain)
```

```
host 192.168.220.130
```

```
# Force record types
host -t A server200.ditiss.com
host -t PTR 192.168.220.130
```

```
#####
```

```
# 9. DEBUGGING (IF REQUIRED)
```

```
#####
```

```
# Check config syntax
```

```
named-checkconf
```

```
# Check zone files
```

```
named-checkzone ditiss.com /var/named/frwd.ditiss.com
```

```
named-checkzone 220.168.192.in-addr.arpa
/var/named/rvs.ditiss.com
```

```
# Logs
```

```
journalctl -xeu named
```

```
#####
```

```
# FINAL RESULT
```

```
#####
```

```
# Forward DNS → Working
```

```
# Reverse DNS → Working
```

```
# Service → Active
```

```
# Permissions → Correct
```

```
#####
```

```

options {
#    listen-on port 53 { 127.0.0.1; }; //this is old conf
#----Enter Your Old Ip----
    listen-on port 53 { 192.168.220.130; };
#----Comment Out V6----
    #listen-on-v6 port 53 { ::1; };
    directory      "/var/named";
    dump-file       "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file   "/var/named/data/named.secrets";
    recursing-file  "/var/named/data/named.recursing";
#----change local host with any----
#    allow-query    { localhost; };
    allow-query     { any; };

```

```

#=====
#=====
//----forward zone----

zone "ditiss.com" IN {
    type master;
    file "frwd.ditiss.com";
};

//----Reverse Zone----

zone "220.168.192.in-addr.arpa" IN {
    type master;
    file "rvs.ditiss.com";
};

```


frwd

```
root@localhost:/var/named - vim frwd.ditiss.com x root@localhost:/var/named

$TTL 1D
@ IN SOA server200.ditiss.com. abc@gmail.com. (
    0      ; serial
    1D     ; refresh
    1H     ; retry
    1W     ; expire
    3H )   ; minimum

@ IN NS server200.ditiss.com.
ditiss.com. IN A 192.168.220.130
www IN A 192.168.220.130
server200 IN A 192.168.220.130
~
~
```

Revers

```
root@localhost:/var/named - vim rvs.ditiss.com x root@localhost:/var/named

$TTL 1D
@ IN SOA server200.ditiss.com. abc@gmail.com. (
    1      ; serial
    1D
    1H
    1W
    3H )

@ IN NS server200.ditiss.com.
130 IN PTR server200.ditiss.com.
~
```

2. Scenario-Based Question (Real Exam Level)

Question:

You are working as a network administrator. A DNS server is configured for the domain **ditiss.com**, but users are facing resolution issues.

Given:

- DNS Server IP: 192.168.220.130
- Domain: ditiss.com
- Host configured: server200.ditiss.com
- Forward and reverse zones are already created

Problem:

When running:

```
host server200.ditiss.com
```

it works 

But:

```
host 192.168.220.130
```

fails 

Also, some clients cannot resolve domain names.

Tasks:

1. Identify possible causes of reverse lookup failure
2. Check and correct reverse zone configuration
3. Verify if the DNS service is properly allowed in the firewall
4. Validate zone files for syntax errors
5. Check network configuration using nmcli
6. Ensure the DNS server is correctly set on client systems
7. Restart required services and test the resolution